

AWS (Amazon Web Services): Comprehensive Theory Study Guide

Module 1: Cloud Computing Basics

- **What is Cloud Computing?** Cloud computing is the on-demand delivery of IT resources (compute, database, storage, networking) over the internet with pay-as-you-go pricing. It eliminates the need to buy, own, and maintain physical data centers.
- **Deployment Models**
 - **Public Cloud:** Owned and operated by third-party cloud providers (e.g., AWS, Azure). Resources are shared with other tenants over the public internet.
 - **Private Cloud:** Cloud infrastructure used exclusively by a single organization. Can be hosted on-premises or by a third-party host.
 - **Hybrid Cloud:** Integrates public cloud services with private clouds or on-premises infrastructure, allowing data and applications to be shared between them.
 - **Multi-Cloud:** Utilizing cloud services from multiple public cloud providers (e.g. AWS and Google Cloud) to prevent vendor lock-in or leverage specific service features.
- **Advantages of Cloud Computing**
 - **Variable Expense:** Trade capital expense (buying hardware) for operational variable expense.
 - **Massive Economies of Scale:** Lower prices due to high-volume aggregate usage.
 - **Stop Guessing Capacity:** Automatically scale resources up or down dynamically.
 - **Agility & Speed:** Provision resources in seconds rather than waiting months for server deliveries.
 - **Focus on Business:** Stop spending money and time maintaining physical server racks.
 - **Go Global in Minutes:** Deploy applications across multiple geographic regions with a few clicks.
- **Disadvantages of Cloud Computing**
 - Downtime risks due to provider network failures.
 - Limited control over low-level hardware configurations.
 - Complexity in managing configurations, which can lead to billing surprises.
 - Data transfer bottlenecks when moving massive datasets out of the cloud.
- **Service Models**
 - **IaaS (Infrastructure as a Service):** Provides raw computing infrastructure (virtual machines, storage, networks). Customer manages OS, runtime, and applications. (e.g., Amazon EC2).

- **PaaS (Platform as a Service):** Provides a managed platform. AWS handles the OS, middleware, and database patching; the customer only deploys application code. (e.g., AWS Elastic Beanstalk).
 - **SaaS (Software as a Service):** A fully completed software application hosted and managed entirely by the provider. (e.g., Salesforce, Microsoft 365).
 - **Serverless Computing:** A cloud execution model where developers write code and the cloud provider handles provisioning, scaling, patching, and resource allocation. Charging is based on active execution millisecond times.
 - **Edge Computing:** Deploying compute resources closer to the source of data generation (end-user devices) rather than routing all queries to centralized cloud datacenters, reducing latency.
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Module 2: AWS Basics

- **What is AWS?** Amazon Web Services is a secure, highly scalable cloud computing platform offering over 200 fully featured services from data centers globally.
 - **AWS Global Infrastructure**
 - **AWS Regions:** Geographic locations consisting of multiple, isolated, and physically separate Availability Zones.
 - **Availability Zones (AZs):** One or more discrete data centers with redundant power, networking, and connectivity in an AWS Region.
 - **Edge Locations:** CDN cache endpoints used by Amazon CloudFront to deliver static content with low latency to nearby users.
 - **AWS Pricing Models**
 - **On-Demand:** Pay-per-second or hour with no upfront commitment.
 - **Savings Plans / Reserved Instances:** Commit to a consistent amount of usage (1- or 3-year term) in exchange for up to a 72% discount.
 - **Spot Instances:** Bid on spare AWS compute capacity for up to a 90% discount. AWS can reclaim instances with a 2-minute warning. Best for stateless, batch jobs.
 - **AWS Free Tier:** Provides newly registered accounts with 12 months of free access to specific services within set usage limits (e.g., 750 hours of t2.micro EC2 monthly).
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Module 3: EC2 (Elastic Compute Cloud)

- **What is EC2?** Elastic Compute Cloud provides secure, resizable virtual servers (instances) in the AWS cloud.
- **EC2 Instance Types** Categorized by optimization profiles: General Purpose (T/M series), Compute Optimized (C series), Memory Optimized (R/X series), Storage Optimized (I/D

series), and Accelerated Computing (G/P GPU series).

- **Dedicated Hosts:** A physical EC2 server fully dedicated to your single use, allowing compliance with specific software licensing rules (BYOL).
 - **AMI (Amazon Machine Image):** A preconfigured template containing the OS, software configurations, and volume structures needed to launch new EC2 instances.
 - **Security Groups** Virtual firewalls acting at the **instance level** to control inbound and outbound traffic. > [!IMPORTANT] > Security Groups are **stateful**. If you allow inbound traffic from an IP, outbound response traffic is automatically allowed, regardless of outbound rules.
 - **Key Pair:** A public and private cryptographic key pair used to securely authenticate SSH access to EC2 instances.
 - **Elastic IP:** A static, public IPv4 address that can be dynamically remapped between EC2 instances to mask server failures.
 - **User Data:** A bootstrap shell script passed during configuration that runs automatically on the instance's very first boot (used for updates, package installs).
 - **Placement Groups**
 - *Cluster:* Places instances in a single rack in one AZ. Low-latency networking.
 - *Spread:* Places instances on physically separate hardware racks across AZs, minimizing core failures.
 - *Partition:* Divides instances into logical partition groups that do not share hardware with other partitions.
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Module 4: Storage

- **Amazon S3 (Simple Storage Service)** Object storage offering high durability (99.999999999% - eleven 9s) and scalability. Files are stored as key-value pairs inside **Buckets**.
 - **S3 Storage Classes:**
 - *S3 Standard:* High durability, availability, and low latency for active data.
 - *S3 Intelligent-Tiering:* Automatically shifts files between access tiers to optimize costs based on usage patterns.
 - *S3 Standard-IA (Infrequent Access):* Cheaper storage but charges retrieval fees. For data accessed less than once a month.
 - *S3 Glacier (Flexible/Deep Archive):* Low-cost archival storage. Retrieval times range from minutes (Glacier) to 12 hours (Deep Archive).
 - **Versioning:** Retains multiple versions of an object in a bucket, protecting against accidental deletions or overwrites.
 - **Lifecycle Policies:** Automatically shifts objects to cheaper storage classes or deletes them after a specified number of days.
 - **Cross-Region Replication (CRR):** Automatically replicates objects from a source bucket to a target bucket in a different AWS region for disaster recovery.

- **S3 Encryption:** Supports Server-Side Encryption (SSE-S3, SSE-KMS, or client-side managed keys).
 - **EBS (Elastic Block Store)** Network block storage volumes attached to EC2 instances. Acts as a raw hard drive. Data is persistent and locked to a single AZ.
 - **EFS (Elastic File System)** A serverless, shared network file system (NFS) that can be mounted simultaneously by hundreds of EC2 instances across multiple Availability Zones.
 - **FSx:** Fully managed third-party file systems (e.g., FSx for Lustre for high-performance computing, FSx for Windows File Server).
 - **Storage Gateway:** Connects on-premises environments to cloud storage (S3) for hybrid cloud architectures.
 - **Comparisons**
 - *EBS vs EFS:* EBS can only be attached to a single EC2 instance at a time in one AZ. EFS can be mounted by hundreds of instances simultaneously across multiple AZs.
 - *EBS vs S3:* EBS is block storage for high-frequency database read/writes. S3 is object storage accessible via HTTP APIs.
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Module 5: IAM (Identity and Access Management)

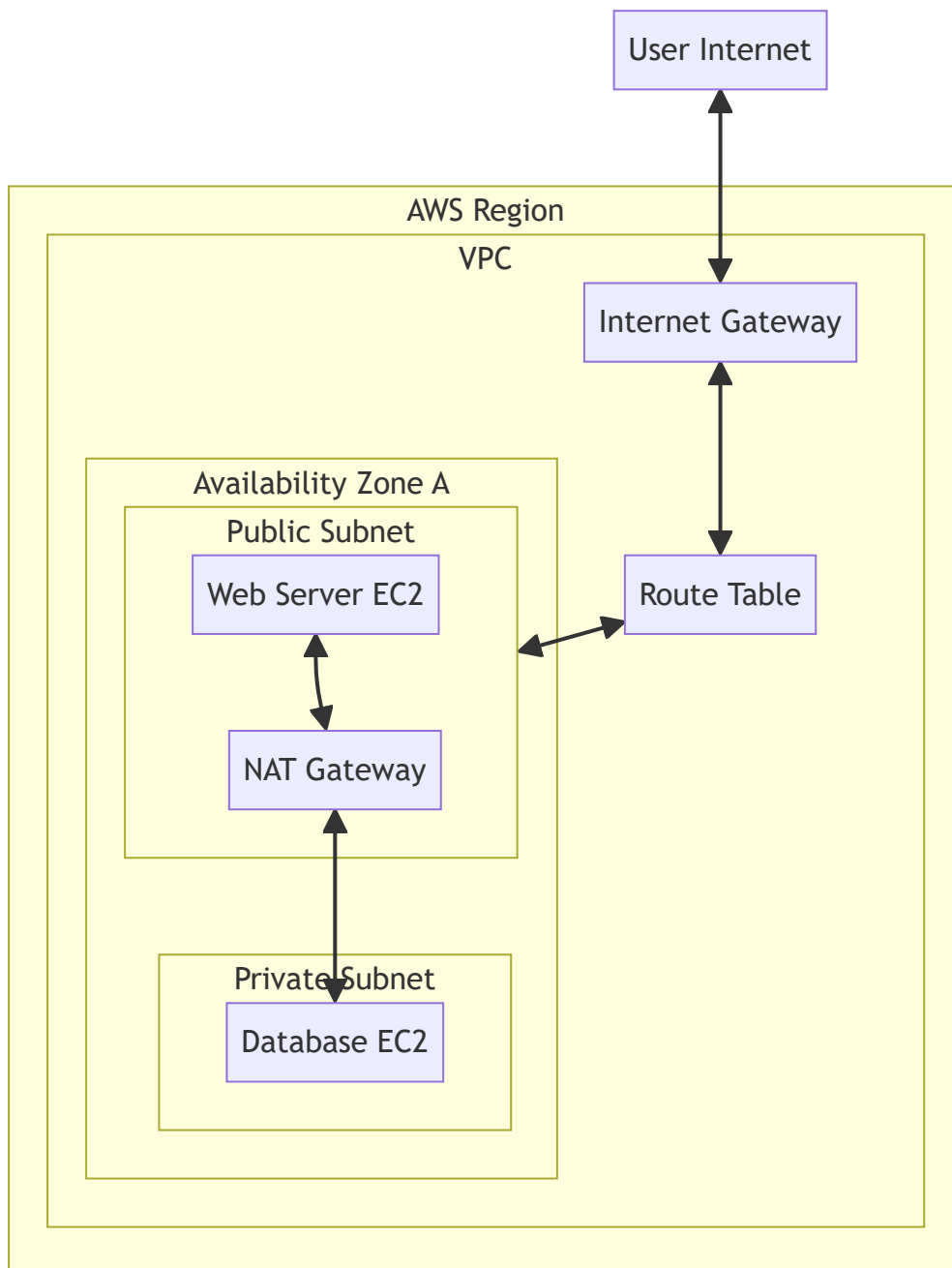
IAM is a global service used to securely control access to AWS resources.

- **Core Components**
 - **IAM User:** A physical person or application identifier.
 - **IAM Group:** A collection of users sharing matching permission privileges.
 - **IAM Role:** An identity assumed by AWS services or users for temporary access. Uses security token services (STS) to generate short-lived credentials.
 - **IAM Policy:** A JSON document defining permissions (Allows or Denies on Action, Effect, Resource).
 - **Managed vs. Inline Policies**
 - *Managed Policies:* Reusable standalone JSON policies managed by AWS or created by customers that can be attached to multiple users, groups, or roles.
 - *Inline Policies:* Hardcoded policies embedded directly inside a single user or role.
 - **Least Privilege Principle:** The security practice of granting users and roles only the minimum permissions necessary to perform their tasks.
 - **MFA (Multi-Factor Authentication):** Recommends adding an extra layer of authentication security alongside standard credentials (username/password).
 - **Temporary Credentials:** IAM roles generate temporary security tokens (STS) that expire automatically, eliminating the need to rotate long-lived credentials.
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Module 6: Networking

1. VPC Architecture Diagram

A Virtual Private Cloud (VPC) provides isolated network environments in the cloud:



2. Networking Concepts

- **VPC (Virtual Private Cloud):** An isolated virtual network within the AWS cloud where you define subnets, route tables, and gateways.
- **Subnets**
 - **Public Subnet:** A subnet containing a route in its Route Table pointing to an Internet Gateway (IGW), allowing bidirectional traffic with the public internet.

- **Private Subnet:** A subnet isolated from the public internet. Contains no route to the IGW.
 - **CIDR (Classless Inter-Domain Routing):** IP address allocation notation (e.g., `10.0.0.0/16` defines a network range of 65,536 IPs).
 - **Internet Gateway (IGW):** A VPC component that enables communication between resources in your public subnets and the internet.
 - **NAT Gateway (Network Address Translation):** A managed gateway placed in a public subnet that allows resources in private subnets to send outbound traffic to the internet, but prevents the public internet from establishing inbound connections.
 - **Security Groups vs. Network ACLs**
 - **Security Groups:** Stateful, acting at the **instance level**. Evaluates all rules before allowing traffic.
 - **Network ACLs (NACL):** Stateless (requires explicit rules for both inbound and outbound traffic), acting at the **subnet level**. Evaluates rules sequentially by number.
 - **Elastic Network Interface (ENI):** A virtual network card that can be attached to EC2 instances.
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Module 7: Load Balancing

- **What is ELB (Elastic Load Balancing)?** A service that automatically distributes incoming application traffic across multiple targets (EC2 instances, containers, Lambdas) in one or more Availability Zones.
 - **Load Balancer Types**
 - **ALB (Application Load Balancer):** Layer 7 (HTTP/HTTPS) load balancer. Supports path-based (`/api`) and host-based (`app.domain.com`) routing.
 - **NLB (Network Load Balancer):** Layer 4 (TCP/UDP) load balancer. Designed for ultra-high throughput and low latency. Can handle millions of requests per second and uses a static IP.
 - **Gateway Load Balancer:** Layer 3 load balancer. Used to deploy, scale, and manage virtual third-party appliances (e.g., firewalls, IDS/IPS).
 - **Health Checks:** Periodic requests sent by the load balancer to target instances. Traffic is routed only to targets that pass health checks.
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Module 8: Auto Scaling

- **What is Auto Scaling?** A service that monitors your applications and automatically adjusts compute capacity up or down to maintain steady, predictable performance at the lowest possible cost.

- **Auto Scaling Group (ASG)** A collection of EC2 instances treated as a logical grouping for scaling and management purposes. You define the Minimum, Maximum, and Desired capacity.
 - **Scaling Policies**
 - **Target Tracking:** Scale resources automatically to keep a metric at a set value (e.g., keep average CPU usage at 60%).
 - **Scheduled Scaling:** Adjusts instance counts at specific times (e.g. scale up before peak business hours).
 - **Dynamic Scaling:** Scale in response to custom CloudWatch alarms (e.g. scale up when SQS queue length grows).
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Module 9: Monitoring

- **Amazon CloudWatch** A monitoring and observability service that collects metrics, monitors logs, and triggers actions based on alarms (e.g. triggering an Auto Scaling action when CPU usage is high).
 - **Amazon CloudTrail** Audits API calls and user activity across the entire AWS account. Logs *who* made a request, *what* action was taken, *when*, and from *where*. Essential for security auditing.
 - **AWS Config:** Tracks configurations and relationships of AWS resources over time, auditing them against compliance rules.
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Module 10: Databases

- **Amazon RDS (Relational Database Service):** Managed relational databases (PostgreSQL, MySQL, SQL Server, Oracle). Handles automated backups, OS patching, and Multi-AZ replication.
- **Amazon Aurora:** A cloud-native, relational database engine (MySQL and PostgreSQL compatible). It is up to 5x faster than standard MySQL, replicates 6 copies of data across 3 AZs automatically, and features self-healing storage.
- **Amazon DynamoDB:** A fully managed, serverless NoSQL database. Delivers single-digit millisecond latency at any scale. Uses partition keys and sort keys for fast retrieval.
- **Amazon Redshift:** Columnar data warehouse designed for high-performance OLAP queries on large datasets.
- **Amazon ElastiCache:** Managed in-memory caching service supporting Redis and Memcached to accelerate application response times.
- **RDS vs. DynamoDB** RDS is structured relational (SQL) for transactional applications requiring ACID joins. DynamoDB is serverless NoSQL for key-value structures that need to scale horizontally with consistent millisecond latencies.

Module 11: Serverless Compute

- **AWS Lambda** An event-driven serverless compute service that executes code in response to triggers. Max execution limit is **15 minutes**.
 - **Lambda Layers**: A mechanism to package common libraries, dependencies, or custom runtimes, making them shareable across multiple Lambda functions to reduce deployment package sizes.
 - **Lambda Triggers**: Events that invoke a Lambda function (e.g., S3 object uploads, DynamoDB streams, API Gateway requests, SQS messages).
 - **API Gateway**: A managed service that allows developers to create, secure, and publish REST/HTTP APIs. Often acts as the entry gate to trigger backend Lambda functions.
 - **Step Functions**: A serverless orchestrator that coordinates multiple Lambda functions and AWS services into workflows using state machines.
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Module 12: Messaging

- **Amazon SQS (Simple Queue Service)** A fully managed, pull-based message queuing service used to decouple application components.
 - *Standard Queue*: Unlimited throughput, guarantees at-least-once delivery, but messages can occasionally arrive out of order.
 - *FIFO Queue*: Guarantees first-in-first-out order and exactly-once processing, but capped at 300 transactions/second (3,000 with batching).
 - **Amazon SNS (Simple Notification Service)** A fully managed, push-based pub/sub messaging service. Fan-out architecture sends messages to multiple subscribers (SQS, Lambda, Email) simultaneously.
 - **SQS vs. SNS** SQS is a queue where one consumer pulls and processes a message. SNS is a pub/sub topic that pushes messages to multiple subscribers simultaneously.
 - **Dead Letter Queue (DLQ)**: A secondary queue where messages that fail processing multiple times are automatically sent for inspection and debugging.
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Module 13: Containers

- **Docker on AWS**: Standard packaging for containerized applications.
- **ECS (Elastic Container Service)**: AWS-native container orchestration service. Very simple and integrates natively with other AWS services.
- **EKS (Elastic Kubernetes Service)**: Managed Kubernetes service, providing standard Kubernetes APIs to deploy and manage containerized applications.

- **Fargate:** A serverless compute engine for containers. Works with both ECS and EKS, eliminating the need to manage EC2 instances for container execution.
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Module 14: Data Engineering Services

- **AWS Glue:** Serverless data integration (ETL) service.
 - **Glue Crawlers:** Scans data in S3 to infer schemas and write tables to the **Glue Data Catalog**.
 - **Athena:** Serverless query service that runs standard SQL queries directly on S3 files.
 - **EMR:** Managed Hadoop/Spark cluster platform for running large-scale data transformations.
 - **Kinesis:** Managed real-time data streaming service (equivalent to Kafka).
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Module 15: Security Services

- **KMS (Key Management Service):** Managed service used to create, control, and rotate cryptographic keys for data encryption.
 - **Secrets Manager vs. Parameter Store**
 - **Secrets Manager:** Stores database credentials and API keys. Supports automatic rotation and integration with KMS.
 - **Parameter Store:** Hierarchical storage for configuration data (strings, passwords). Free tier available; does not support automated rotation.
 - **WAF (Web Application Firewall):** Protects web applications against SQL injections, cross-site scripting (XSS), and common web exploits.
 - **Shield:** DDoS protection service. Standard is free; Advanced offers dedicated DDoS response team support.
 - **Cognito:** User directories and access control for web and mobile applications.
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Module 16: DevOps

- **CodePipeline:** Orchestrates CI/CD workflows from source code to deployment.
 - **CodeBuild:** Serverless build and testing service.
 - **CodeDeploy:** Automates deployment tasks to EC2, Fargate, or on-premises servers.
 - **CloudFormation:** Infrastructure as Code (IaC) service that provisions AWS resource stacks using declarative YAML or JSON templates.
 - **Elastic Beanstalk:** Platform as a Service (PaaS). Automates capacity provisioning, load balancing, and scaling; developers only deploy code.
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Module 17: AI & ML Services

- **Amazon SageMaker:** A comprehensive platform to build, train, tune, and deploy machine learning models. Provides hosted endpoints for real-time model inference.
 - **AI Application Services**
 - *Textract:* Automatically extracts text and tabular data from documents.
 - *Rekognition:* Image and video analysis (object and face detection).
 - *Comprehend:* NLP service that extracts insights and sentiments from text.
 - *Lex:* Conversational AI for building voice and text chatbots.
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Module 18: GenAI on AWS

- **Amazon Bedrock** A fully managed serverless service that offers access to leading Foundation Models (FMs)—such as Anthropic Claude, Meta LLaMA, Mistral, and Amazon Titan—through a single unified API.
 - **Knowledge Bases:** Managed RAG workflows in Bedrock. Automatically manages document indexing, chunking, embedding generation, and vector database updates in OpenSearch Serverless.
 - **Bedrock vs. SageMaker** Bedrock is a serverless API for pre-trained foundation models. SageMaker is used to build, train, and host custom machine learning models from scratch.
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Module 19: High Availability & DR

- **Multi-AZ:** Synchronous replication to a standby database or server in a different AZ, enabling automated failover in case of a datacenter outage.
 - **Multi-Region:** Replicating architectures across separate geographic regions for disaster recovery and low-latency global user access.
 - **RPO vs. RTO**
 - **RPO (Recovery Point Objective):** The maximum amount of data loss an organization can tolerate, measured in time (e.g. maximum 1 hour of lost transactions).
 - **RTO (Recovery Time Objective):** The maximum allowable downtime before systems must be restored after an outage.
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Module 20: Performance Optimization

- **CloudFront:** Content Delivery Network (CDN) that caches web content at globally distributed Edge Locations to reduce latency.

- **Route53:** Domain Name System (DNS) web service. Supports routing policies such as latency-based, geolocation, failover, and weighted routing.
 - **Cost Optimization:** Implementing S3 Lifecycle policies to archive old files, using Spot instances for batch processing, purchasing Savings Plans, and right-sizing EC2 instances.
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Module 21: Scenario-Based Questions

1. How would you deploy a FastAPI application on AWS?

Package the FastAPI application as a Docker image. Push the image to **Amazon ECR**. Create an **ECS Fargate** task definition. Deploy it behind an **Application Load Balancer (ALB)** to distribute traffic and auto-scale containers based on request count.

2. How would you host a RAG application?

Store document PDFs in **S3**. Set up **Amazon Bedrock Knowledge Bases** to load the documents, chunk them, embed them, and index them into an **Amazon OpenSearch Serverless** vector index. Query models like Anthropic Claude via the Bedrock API.

3. How would you secure an AWS application?

Isolate resources in a **VPC**. Place databases in **private subnets**. Configure **Security Groups** to restrict access to instances. Store database credentials in **Secrets Manager**, and enable **WAF** on the load balancer to protect against web exploits.

Module 22: Architecture & Best Practices

1. AWS Well-Architected Framework

Provides design principles and architectural guidelines across 6 pillars: 1. **Operational Excellence:** Automating changes, monitoring health, and continuously improving procedures. 2. **Security:** Protecting data, systems, and assets using IAM, encryption, and network isolation. 3. **Reliability:** Designing workloads to recover from infrastructure failures automatically (horizontal scaling, fault tolerance). 4. **Performance Efficiency:** Right-sizing resources and selecting optimized storage/compute options. 5. **Cost Optimization:** Eliminating unneeded resources and utilizing compute savings models (Spot, Savings Plans). 6. **Sustainability:** Minimizing environmental impacts and maximizing utilization. * **Shared Responsibility Model** * **AWS:** Security *of* the cloud (global infrastructure, physical data centers, host OS hypervisors). * **Customer:** Security *in* the cloud (guest OS patching, firewall settings, data encryption, IAM policies).

★ Top 40 Most Frequently Asked AWS Questions (Accenture)

1. **What is AWS?** Amazon Web Services: A secure cloud computing platform offering over 200 fully featured services globally.
2. **IaaS vs PaaS vs SaaS?**
 - IaaS: Infrastructure (virtual machines, networks). User manages OS.
 - PaaS: Managed platforms (OS and database managed by provider). User deploys code.
 - SaaS: Fully completed hosted software apps.
3. **Regions vs Availability Zones?** A Region is a separate geographic area containing multiple AZs. An AZ consists of one or more physically isolated datacenters.
4. **Describe S3 Storage Classes.** Standard (active access), Standard-IA (infrequent access), Intelligent-Tiering (auto-shifts based on use), and Glacier (archive).
5. **EBS vs EFS vs S3?**
 - EBS: Network block storage attached to a single EC2 in one AZ.
 - EFS: Shared network file system mountable by multiple instances.
 - S3: HTTP-based scalable object storage.
6. **What is a VPC?** Virtual Private Cloud: An isolated virtual network inside your AWS account.
7. **Public vs Private Subnet?** Public subnets contain a route pointing to an Internet Gateway. Private subnets do not, isolating them from public internet access.
8. **Security Group vs NACL?** Security groups are stateful and act at the instance level. NACLs are stateless, processed sequentially, and act at the subnet level.
9. **Internet Gateway vs NAT Gateway?** An IGW enables public subnet resources to communicate with the internet. A NAT Gateway allows private subnet resources to reach the internet without exposing them to inbound connections.
10. **What is AWS Lambda?** A serverless compute service that executes code in response to triggers, with a 15-minute maximum run limit.
11. **Lambda vs EC2?** Lambda is serverless, auto-scales instantly, and is priced per millisecond of execution. EC2 provides virtual servers where you manage the OS, paying for the uptime of the server.
12. **What is API Gateway?** A serverless service that allows you to create, secure, and publish REST/HTTP APIs.
13. **Explain Auto Scaling.** Dynamically adjusts EC2 instance counts inside an Auto Scaling Group to match application load.
14. **What is an Application Load Balancer (ALB)?** A Layer 7 load balancer that routes traffic to targets based on path or host headers.
15. **CloudWatch vs CloudTrail?** CloudWatch monitors metrics, logs, and triggers system alarms. CloudTrail audits API calls and user activities across the AWS account.
16. **RDS vs DynamoDB?** RDS is structured relational SQL. DynamoDB is managed serverless key-value NoSQL offering consistent millisecond latencies.

17. **What is Amazon Redshift?** A petabyte-scale columnar data warehouse optimized for OLAP.
18. **AWS Glue vs Athena?** Glue is an ETL integration service containing crawler and Spark engines. Athena is a serverless query service running SQL queries on S3 files.
19. **SQS vs SNS?** SQS is a pull-based queue for one consumer. SNS is a push-based pub/sub topic for fan-out messages to multiple subscribers.
20. **What is Fargate?** A serverless compute engine for containers that eliminates the need to manage EC2 instances for ECS or EKS.
21. **What is AWS KMS?** Key Management Service: A service used to create, rotate, and manage cryptographic keys.
22. **Secrets Manager vs Parameter Store?** Secrets Manager handles credentials, integrates with KMS, and supports automatic rotation. Parameter Store is a configuration store without auto-rotation.
23. **What is Amazon Bedrock?** A fully managed serverless service offering API access to leading foundation models.
24. **Multi-AZ vs Multi-Region?** Multi-AZ provides datacenter redundancy and failover in the same region. Multi-Region provides disaster recovery and low-latency global user routing.
25. **What is the Well-Architected Framework?** A set of design principles across 6 pillars: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, and Sustainability.
26. **Explain the Shared Responsibility Model.** AWS is responsible for security *of* the cloud. The customer is responsible for security *in* the cloud (data, configurations, IAM).
27. **What is CloudFormation?** An IaC service that provisions resource stacks using YAML/JSON templates.
28. **What is AWS Glue Data Catalog?** A Hive-compliant metadata repository for structured tables stored on S3.
29. **What is Amazon EMR?** A managed platform for running big data frameworks like Apache Spark and Hadoop.
30. **Explain Blue-Green Deployment.** Creating two identical environments: Blue (active production) and Green (new release). Traffic is swapped to Green when verified, reducing risk.
31. **What is Amazon Route53?** A highly available, scalable Domain Name System (DNS) web service.
32. **What is AWS Shield?** A managed DDoS protection service. Standard is free; Advanced offers dedicated DDoS response team support.
33. **What is AWS Config?** A service that records resource configurations, auditing compliance rules.
34. **What is Amazon Cognito?** A service that provides sign-up, sign-in, and access control for web and mobile applications.
35. **What are S3 Lifecycle Policies?** Automated rules that transition objects to cheaper storage classes or delete them after a set period.
36. **Explain Elastic IP.** A static public IPv4 address that can be remapped dynamically between instances to mask failures.

37. **What is a Dead Letter Queue (DLQ)?** A queue where messages that fail processing multiple times are sent for debugging.

38. **What is Kinesis?** A managed real-time data streaming service.

39. **What is AWS Organizations?** A service that allows you to consolidate and manage multiple AWS accounts centrally.

40. **End-to-End ML Architecture on AWS**

1. Data stored in **S3**.
2. ETL performed using **AWS Glue** or **EMR** to register tables in the **Glue Data Catalog**.
3. Model built, trained, and tuned in **Amazon SageMaker**.
4. Model hosted as a SageMaker real-time endpoint.
5. fastAPI app on **ECS Fargate** routes user requests to SageMaker endpoint.
6. Monitored with **CloudWatch** and audited with **CloudTrail**.



Module Summary & Key Takeaways

Theory Focus: Virtual Private Clouds (VPCs), Managed Database Services, Serverless, and Decoupled Queuing.

This module covers key services and design frameworks of Amazon Web Services. Essential learnings include:

- **VPC Isolation & Networking:** Private cloud boundaries are structured using Subnets, Route Tables, and Internet Gateways (IGW). Public instances communicate bidirectionally with the internet, while private database instances remain isolated, routing outbound requests securely through NAT Gateways.
- **Managed Compute & Storage:** Compute options range from static virtual instances (EC2) to serverless runtime functions (**AWS Lambda**). Storage is managed across Object (S3), Relational (RDS), and serverless NoSQL (DynamoDB) databases.
- **Asynchronous Decoupling:** Architectures remain fault-tolerant and scale horizontally by decoupling systems using **Simple Queue Service (SQS)** for message buffering and **Simple Notification Service (SNS)** for publish-subscribe messaging.